



dart-practice-lab

Visitors

My Dart learning journey including Lists, Sets, Maps, OOP, Problem Solving, and Mini Projects.

Dart

List

Set

Map



Dart List, Set & Map Practice Problems



Lab Topic: Collection Framework (List, Set, Map)



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1. Sum of List Elements

Question

Create a list of integers and calculate the sum of all elements.

Task

- Create a list of numbers.
 - Calculate the total sum.
 - Print the result.
-

Solution

```
void main() {  
    List<int> numbers = [10, 20, 30, 40];  
  
    int sum = 0;  
  
    for (int num in numbers) {  
        sum += num;  
    }  
  
    print("Sum = $sum");  
}
```

Output

Sum = 100

2. Find Largest Number

Question

Take a list of numbers and print the largest number.

Task

- Traverse the list.
 - Find the maximum value.
 - Print the result.
-

Solution

```
void main() {  
    List<int> numbers = [10, 50, 25, 80, 40];  
  
    int largest = numbers[0];  
  
    for (int num in numbers) {  
        if (num > largest) {  
            largest = num;  
        }  
    }  
  
    print("Largest Number = $largest");  
}
```

Output

Largest Number = 80

3. Reverse a List

Question

Reverse a list without using the built-in `reversed()` method.

Task

- Create a new list.
- Add elements from the end of the original list.

Solution

```
void main() {  
    List<int> numbers = [1, 2, 3, 4, 5];  
  
    List<int> reversedList = [];  
  
    for (int i = numbers.length - 1; i >= 0; i--) {  
        reversedList.add(numbers[i]);  
    }  
  
    print(reversedList);  
}
```

Output

```
[5, 4, 3, 2, 1]
```

4. Count Even and Odd Numbers

Question

Count how many even and odd numbers exist in a list.

Solution

```
void main() {  
    List<int> numbers = [1, 2, 3, 4, 5, 6];
```

```
int even = 0;
int odd = 0;

for (int num in numbers) {
    if (num % 2 == 0) {
        even++;
    } else {
        odd++;
    }
}

print("Even = $even");
print("Odd = $odd");
}
```

Output

```
Even = 3
Odd = 3
```

5. Remove Duplicate Values

Question

Remove duplicate elements from a list.

Solution

```
void main() {
    List<int> numbers = [1, 2, 2, 3, 4, 4, 5];

    List<int> unique = numbers.toSet().toList();

    print(unique);
}
```

Output

```
[1, 2, 3, 4, 5]
```

6. Second Largest Element

Question

Find the second largest number in a list.

Solution

```
void main() {  
    List<int> numbers = [10, 40, 20, 80, 60];  
  
    numbers.sort();  
  
    print(numbers[numbers.length - 2]);  
}
```

7. Merge Two Lists

Solution

```
void main() {  
    List<int> a = [1, 2, 3];  
    List<int> b = [3, 4, 5];  
  
    List<int> merged = {...a, ...b}.toList();  
  
    print(merged);  
}
```

8. Search an Element

Solution

```
void main() {  
    List<int> numbers = [10, 20, 30, 40];
```

```
    print(numbers.contains(30));  
}
```

9. Sort List in Ascending Order

Solution

```
void main() {  
    List<int> numbers = [5, 2, 8, 1, 9];  
  
    for (int i = 0; i < numbers.length; i++) {  
        for (int j = i + 1; j < numbers.length; j++) {  
            if (numbers[i] > numbers[j]) {  
                int temp = numbers[i];  
                numbers[i] = numbers[j];  
                numbers[j] = temp;  
            }  
        }  
    }  
  
    print(numbers);  
}
```

10. Rotate List

Solution

```
void main() {  
    List<int> numbers = [1, 2, 3, 4, 5];  
  
    int first = numbers.removeAt(0);  
  
    numbers.add(first);  
  
    print(numbers);  
}
```

11. Create a Set

```
void main() {
    Set<String> fruits = {
        "Apple",
        "Banana",
        "Mango"
    };

    print(fruits);
}
```

12. Check Unique Elements

```
void main() {
    List<int> numbers = [1, 2, 3, 4];

    bool isUnique =
        numbers.length == numbers.toSet().length;

    print(isUnique);
}
```

13. Union of Two Sets

```
void main() {
    Set<int> a = {1, 2, 3};
    Set<int> b = {3, 4, 5};

    print(a.union(b));
}
```

14. Intersection of Sets

```
void main() {  
    Set<int> a = {1, 2, 3};  
    Set<int> b = {2, 3, 4};  
  
    print(a.intersection(b));  
}
```

15. Difference Between Sets

```
void main() {  
    Set<int> a = {1, 2, 3, 4};  
    Set<int> b = {3, 4};  
  
    print(a.difference(b));  
}
```

16. Remove Duplicate Characters

```
void main() {  
    String word = "programming";  
  
    print(word.split('').toSet().join());  
}
```

17. Find Common Friends

```
void main() {  
    Set<String> user1 = {  
        "Rony",  
        "Sakib",  
        "Nayeem"  
    };  
}
```

```
Set<String> user2 = {  
    "Sakib",  
    "Nayeem",  
    "Hasan"  
};  
  
print(user1.intersection(user2));  
}
```

18. Subset Checking

```
void main() {  
    Set<int> a = {1, 2};  
    Set<int> b = {1, 2, 3, 4};  
  
    print(b.containsAll(a));  
}
```

19. Student Marks System

```
void main() {  
    Map<String, int> marks = {  
        "Rony": 85,  
        "Sakib": 92,  
        "Nayeem": 78  
    };  
  
    print(marks);  
}
```

20. Word Frequency Counter

```
void main() {  
    String text = "dart is easy dart is powerful";  
  
    List<String> words = text.split(" ");  
}
```

```
Map<String, int> frequency = {};  
  
for (String word in words) {  
    frequency[word] =  
        (frequency[word] ?? 0) + 1;  
}  
  
print(frequency);  
}
```

21. Find Highest Value

```
void main() {  
    Map<String, int> marks = {  
        "Rony": 85,  
        "Sakib": 95,  
        "Nayeem": 78  
    };  
  
    String topper = "";  
    int highest = 0;  
  
    marks.forEach((name, mark) {  
        if (mark > highest) {  
            highest = mark;  
            topper = name;  
        }  
    });  
  
    print("$topper : $highest");  
}
```

22. Check Key Exists

```
void main() {  
    Map<String, int> marks = {  
        "Rony": 85  
    };  
}
```

```
    print(marks.containsKey("Rony"));  
}
```

23. Merge Two Maps

```
void main() {  
    Map<String, int> map1 = {  
        "A": 1  
    };  
  
    Map<String, int> map2 = {  
        "B": 2  
    };  
  
    Map<String, int> merged = {  
        ...map1,  
        ...map2  
    };  
  
    print(merged);  
}
```

24. Character Frequency Counter

```
void main() {  
    String text = "hello";  
  
    Map<String, int> frequency = {};  
  
    for (String ch in text.split('')) {  
        frequency[ch] =  
            (frequency[ch] ?? 0) + 1;  
    }  
  
    print(frequency);  
}
```

25. Phone Book Application

```
void main() {
    Map<String, String> phoneBook = {};

    phoneBook["Rony"] = "01700000000";

    print(phoneBook["Rony"]);

    phoneBook["Rony"] = "01800000000";

    phoneBook.remove("Rony");

    print(phoneBook);
}
```

26. Employee Database

```
void main() {
    Map<int, String> employees = {
        101: "Rony",
        102: "Sakib"
    };

    employees[103] = "Nayeem";

    print(employees[101]);

    employees[102] = "Hasan";

    employees.remove(101);

    print(employees);
}
```

Learning Outcomes

After completing this lab, you will be able to:

- Work with Dart Lists
- Use Sets for unique data
- Store data using Maps
- Count frequencies
- Search and sort data
- Remove duplicates
- Perform collection operations
- Solve basic interview questions

★ If this repository helps you, don't forget to star it.